

NOVEL FEATURE GENERATION AND CLASSIFICATION FOR A 2-STATE SELF-PACED BRAIN COMPUTER INTERFACE SYSTEM

Jia Gu¹, Rabab Ward¹

¹Department of Electrical and Computer Engineering, University of British Columbia

ABSTRACT

A two-state Self-paced Brain Computer Interface (SBCI) system enables users to control external devices at any time they desire by intentionally switching their mental states. In order to accurately control devices, distinguishing features representing different brain states must be present in the EEG signals. This paper introduces a novel feature generation method for EEG motor imagery data, based on Multivariate Empirical Mode Decomposition (MEMD), the Hilbert Transform and a phase synchronization index. Novelties of our approach are (1) MEMD is applied for decomposing an EEG signal into its narrow-band frequency components, from which features of the different brain states are calculated, and (2) a phase synchronization index is calculated without averaging over trials or time. We used a simple and fast classification scheme that employed an empirical threshold value obtained from the Receiver Operating Characteristic (ROC) curve of the training data. Applying the proposed method on only two mono-polar EEG channels, the SBCI system with the selected features yields a 93% True Positive rate, and a 5.8% False Positive rate using the BCI competition III data set Iva.

Index Terms — self-paced BCI, MEMD, phase synchronization, EEG, motor imagery

1. INTRODUCTION

Most existing BCIs are synchronous, i.e. they allow users to operate the BCI at specific times dictated by the system. A Self-paced Brain Computer Interface (SBCI) system, however, enables humans to control external devices whenever they want. SBCIs enable those who have lost their motor control abilities to interact with the external world at their own pace. Thus, it is critical to have a robust and accurate SBCI system with a low false positive rate in order to reduce user frustration. To operate any BCI, changes in the user's brain states are detected from electroencephalography (EEG) measurements. For SBCIs, the brain state during which the user is performing a specific imagined movement, such as a hand movement, is called the intentional control (IC) state. During any other time when the user does anything else but performing the imagined hand movement, is denoted as the no-control (NC) state. The accuracy of a SBCI system can be quantified in terms of true positive (TP) and false positive (FP) rates. The

classification accuracy of a BCI system results directly from the system's ability to generate and select distinguishing IC and NC features from EEG signals. EEG signals are known to be extremely high dimensional, non-stationary, noisy, and also suffer from the volume conduction effect. The focus of this paper is to introduce a novel feature generation method to distinguish IC from NC states, with the objective of maximizing accuracy. The proposed system generates features using Multivariate Empirical Mode Decomposition (MEMD), the Hilbert Transform (HT), and a 1:1 phase synchronization index based on Shannon entropy. MEMD decomposes the EEG signals into components whose frequencies are narrow banded. In the case of multiple EEG channels, MEMD guarantees the same number of components per channel, and the corresponding components across all channels have the same frequency range. Then, the HT is applied on each component to calculate the instantaneous frequencies. Lastly, for each second, the 1:1 phase synchronization index is calculated from two corresponding components of two EEG channels. This index reflects the pair-wise synchronization level, and is used to classify IC vs. NC states.

In this study, the data were gathered from five healthy subjects performing imagined right hand movement [1], and evaluated in a self-paced fashion to yield one decision per second.

Our method is compared with those in [21, 22] using the same data set. While our work is designed for a self-paced system, all the published work on this data set were designed for synchronous systems. To compare, we replaced our feature generation and classification blocks by those in [21, 22], where the Common Spatial Pattern (CSP) was used on different frequency bands of channels C3 and C4 for feature extraction, and the support vector machine (SVM) and the linear discriminate analysis (LDA) algorithms were used for classification respectively. The comparison results are recorded in Table 2.

2. PROPOSED APPROACH

2.1 Reasons for choosing phase synchronization indices as features

2.1.1 We chose features based on the synchronization or coupling of the EEG channels as done in [7]. The brain can be described as a complex network of coupled nonlinear and stochastic dynamical subsystems which are called functional sources [2]. Nonlinear measures obtained from a single time

series (mono-polar or bi-polar EEG channel measurements) describe the lower level functional sources, while measures of coupling or synchronization between time series describe the higher level of functional network. Thus, [2] concludes that “measures of coupling in functional networks may be more reliable estimates of synchronization levels than local estimates of dynamical complexity”. Synchronization was found to not only occur between regular, linear oscillating systems, but also between irregular, chaotic systems [3], which make it a more suitable measure to analyze EEG signals. Also, based on animal studies done in [4], it was discovered that the control of coordinated movements is based on a synchronization of the firing activity of groups of neurons in the primary and secondary motor areas.

2.1.2. We chose a nonlinear synchrony measure to quantify the synchronization levels of the EEG channels as in [7] because the human brain cannot be modeled as a linear system.

2.1.3. We chose phase synchronization to quantify synchrony level among EEG channels, as in [7]. Studies in [6] showed that an internally paced motor imaginary event generates event-related potentials that are phase-locked to the event. Phase synchronization measures interdependencies of the instantaneous phases of two time series at every time point, which is independent of the amplitudes of the signals [5]. It was chosen to identify the event occurrence because it is more suitable to track non-stationary signals generated from nonlinear processes.

2.2 Commonly used algorithms to extract phase information and their corresponding drawbacks

Classical phase measures can be calculated from the HT [8], or from a time-varying complex energy spectrum. This spectrum could be obtained from the wavelet algorithm with a Morlet wavelet [9] or the short-time Fourier transform (STFT) [10]. The Fourier transform suffers limitations that the system must be linear and the data must be periodic and stationary [11]. Most EEG analysis based on the Fourier transform use the STFT obtained by a sliding window of a short length. The signals within this window are considered to be piece-wise stationary. But it has been argued [11] that it is hard to guarantee that the use of a fixed window size corresponds to stationary time scales of the signals under study. Wavelet analysis, pointed out by [11], “is basically a linear analysis”. Another problem is that the selected basis wavelet is used to analyze all the data, thus making wavelet analysis non-adaptive. The HT can only produce a meaningful instantaneous phase when applied to signals composed of narrow bands of frequencies. Normally one

has to band-pass the signals of interest to the desired frequency ranges before applying the HT. Filters normally suffer from two major drawbacks: (1) lacking precise cut-offs, and (2) it is difficult to find a filter with excellent resolution in both time and frequency domain so as to preserve the phase information [12]. The HT is more desirable compared to STFT and wavelet because it is parameter free. The aforementioned drawbacks vanish immediately when Empirical Mode decomposition (EMD) combined with the HT [11] are applied to obtain instantaneous phase measures, because algorithms based on this combination are designed for non-stationary signals produced by nonlinear systems.

2.3 EMD and its algorithms

The frequencies of non-stationary data change from instance to instance, giving rise to a need to calculate local, or instantaneous frequencies. The Hilbert Transform $Y(t)$ of an arbitrary time series $X(t)$, is defined as: $Y(t) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{X(t')}{t-t'} dt'$, where P is the Cauchy principal value. $\theta(t) = \arctan\left(\frac{Y(t)}{X(t)}\right)$, and the instantaneous frequency is controversially defined as $w = \frac{d\theta(t)}{dt}$. In order for an instantaneous frequency to make sense, the condition of narrow banded frequency is imposed. There are two physical requirements needed for a narrow band signal to be defined: (1) the total number of extrema and zeros crossings must differ by at most one in the entire dataset and (2) at any given point, the mean value of the envelopes defined by its local maxima and minima equals to zero. The intrinsic mode functions (IMF), for which instantaneous frequencies can be defined everywhere, are invented to qualify for these two requirements [11].

EMD adaptively decomposes any signal into a finite collection of IMFs via a sifting algorithm based on the signal's local time scale characteristics. Each obtained IMF represents a narrow frequency band of the original signal, without any complex riding waves. This makes EMD a good solution to non-stationary and nonlinear time series analysis due to its adaptivity, locality, completeness, and almost orthogonal basis. The study in [13] has confirmed improvements in time-frequency resolution using EMD compared to wavelet or STFT. In [14], it is pointed out that the high synchronization index obtained from superimposed EEG signals might not reflect true synchronization. Thus in [14] Independent Component Analysis is first applied to transform the EEG signals into reliable basis before calculating the synchronization indices. Applying EMD to the original signal automatically bypass this problem caused by the superposition of signals. The high temporal

resolution of EEG also enhances the accuracy of the EMD algorithm. The HT is then applied to each individual IMF to obtain the instantaneous phase.

2.4 MEMD and its algorithm

EMD can only be applied to one EEG channel at a time. Thus it does not guarantee that two EEG channels will have the same number of frequency band decompositions. This makes phase synchrony evaluation between two signals at a matching frequency band impossible [17]. The MEMD algorithm does not have this limitation as it has been developed to simultaneously extract the same IMF components for 3 to 16 channels [18]. The algorithm is as follows:

1. Choose a suitable point set for sampling on an (n-1) sphere
2. Calculate a set of projection ($p^{0k}(t)|t=1:T$) of the input signal, along the whole set of direction vectors
3. Locate time instants (t^{0k}) corresponding to the maxima in $p^{0k}(t)|k=1:k$
4. Interpolate using t^{0k} and the input signals to obtain a set of multivariate envelopes $e^{0k}(t)|k=1:k$.
5. Calculate the mean of the envelopes by: $m(t) = \frac{1}{K} \sum_{k=1}^K e^{0k}(t)$
6. Extract the detail $d(t) = x(t) - m(t)$. If $d(t)$ is an IMF, repeat above steps to $x(t) - d(t)$, otherwise apply them to $d(t)$.

2.5 The m:n phase locking index to measure phase synchronization

Phase synchrony between two frequencies can be defined as the ability of the phase difference, Φ_{12} , to stay bounded over time, i.e. $\Phi_{12}(t) = |n\Phi_1(t) - m\Phi_2(t)| < \text{constant}$, where $\Phi_1(t)$ and $\Phi_2(t)$ are the instantaneous phases of two time series at time instance t. An extensive EEG phase synchronization literature survey finds the phase locking value [15] to be the single most dominant measure used. It is defined as:

$PLV = |\langle \exp(j\Phi_{12}(t)) \rangle|$, where $\langle \bullet \rangle$ is the averaging operator over many trials. Despite its popularity and robustness, this measure requires taking averages over trials, which limits its application for real-time/online EEG classification. We used the 1:1 phase locking index proposed by Tass *et al.* [16]. They developed the m:n phase locking index to quantify the strength of synchronization of bivariate non-stationary time series based on Shannon entropy. The distribution of the cyclic relative phase can be expressed as: $\beta_{n,m}(t) = (n\Phi_1(t) - m\Phi_2(t)) \bmod 2\pi$, where n

and m are the order of the locking (chosen empirically), $\bmod 2\pi$ is used for normalization. The 1:1 phase synchronization index used in this study is defined as: $\gamma = (S_{\max} - S)/S_{\max}$, where $S = -\sum_{k=1}^N p_k \ln p_k$ is the entropy of the distribution of $\beta_{n,m}$, and $S_{\max} = \ln N$, where N is the number of bins used. $\gamma=0$ corresponds to no synchronization at all, and $\gamma=1$ corresponds to perfect synchronization.

3. PERFORMANCE EVALUATION AND RESULTS

3.1 Experimental settings

The data used [1] was down-sampled from 1000Hz to 250 Hz, then low pass filtered to be below 30 Hz. The data starting two seconds prior till five seconds after the cue onsite were taken per trial. MEMD was applied to 15 of the 118 channels, covering the supplementary motor area and the sensory-motor cortex. Mono-channels C3 and C4 are used not only because they correspond to the primary sensor motor hand representation area, but also because the center-to-center electrode separation distance between them is about 8cm, which greatly reduces the volume conduction effect [19]. The C3-C4 pair also yields the best empirical performance out of all pair combinations. Then the HT was applied on each IMF to obtain the instantaneous phases. The IFMs generated are ranked from highest to lowest frequencies. The highest frequency IMF component is normally associated with muscle artefacts [20], and is disregarded. The last IMF is the mean trend, and produces no useful information. The frequency range of the second to last IMF is too low to give any distinguishing information, and thus is disregarded. For each of the instantaneous phase vectors obtained for each remaining IMF component, we obtain a phase synchronization index every second.

3.2 Simple classification scheme based on a threshold value

An IC period is declared as the output when a significant change (based on the threshold value selected from the training data's ROC) in phase synchronization indices between two consecutive windows is observed. For each of the five subjects, the 140 trials were randomly divided into 50% training and 50% testing data. The threshold values were incremented by 0.01 at a time from 0 to 1 to generate the training data's ROC curve. The threshold value that corresponds to the maximum TP and its corresponding minimum FP is selected (3.2.1). This threshold value is then used to evaluate the testing data of each trial. This entire evaluation procedure is then repeated over 100 times, and the average TP and FP per subject is obtained, and recorded in Table 1.

3.2.1 Criteria of maximizing TP while minimizing FP

The threshold that yields the largest TP value and its corresponding minimum FP value is used unless the difference between the 1st and 2nd largest TP values is small, yet the corresponding FP values differ by a factor of 1.5 or greater, in which case the threshold value corresponding to the 2nd largest TP is used.

3.3 Results

Table 1. Classification results obtained from our proposed approach

Subjects	Average TP	Average FP	TP std	FP std
1	95.66%	2.85%	1.78	1.58
2	94.73%	5.1%	2.39	1.54
3	92.98%	7.24%	3.88	2.05
4	90.60%	6.13%	5.42	2.13
5	90.99%	7.86%	4.43	1.60
Average	93%	5.83%	3.58	1.78

Table 2. Comparison with commonly used approaches on same data set

	Average TP	Average FP
Proposed approach	93%	5.83%
CSP+LDA	48.05%	28.3%
CSP+SVM	99.8%	54.7%

5. DISCUSSION AND CONCLUSION

It is evident from the results presented in Table 1 and 2 that our approach produces a high average TP rate at a significantly lower FP value. For a SBCI system, it is critical to reduce the FP value to minimize user frustration. The FP rate of 5.8% corresponds to only 3 FPs per minute. In summary, the combination of MEMD and the HT algorithms is a natural solution to process non-stationary EEG signals generated from nonlinear sources without violating any fundamental EEG signal properties. Our proposed novel feature generation method produced good results using only two EEG channels. Introducing MEMD for EEG signal decomposition allowed us to naturally obtain more accurate and meaningful instantaneous phase components, which are then used to calculate and compare synchronization indices across frequency bands of different channels. This has enhanced the accuracy of IC detections while minimizing the FP rate. For future work, a combination of features can be used, such as combining phase synchronization value with changes in power or including more channels.

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